



NSCA
National Strength and Conditioning Association

CHAPTER SIX | SAFE TRAINING

Safe training involves “risk management,” which is the modern term for ensuring that programs, policies, procedures, personnel, and other factors are selected and act in the safest manner possible. Risk management is the modern term because absolute safety remains unachievable; thus, the coach must “manage” risk such that the probability of injury is low. “Injury prevention” is also a common term for safe training in that the obvious goal is to prevent injury. However, modern understanding of safety programs and their implementation involves management more than an implied guarantee of prevention. Therefore, the following information should be used to develop and implement sound strength training and conditioning programs, and facility policies and procedures, to manage risk effectively and help ensure a safe training environment.

Waivers and Informed Consent/Assent

1. Informed consent requirements now permeate many activities from school field trips to serious medical procedures. Every athlete should be made fully aware of the risks involved in the activity and formally (in writing) agree to undertake those risks.
2. Consent is a term used for people of legal age who are able to sign a legally binding contract. Assent refers to the agreement of the underage participant to take part in the activities and signifies that the underage participant understands the risks involved. Underage athletes will also require the consent of their parent or legal guardian.
3. “Technically, informed consent is a contract.” Both parties in the contract must understand the agreement they are making. These types of contracts cannot legally bind minors, because of their immaturity. In order for minors to be involved, their parents and/or guardians must agree to the contents of the document. Informed consent should always be obtained in writing (45).
4. Informed consent is achieved by acknowledging risk, developing a document that details these risks, presenting the document, and ensuring complete understanding of the risks by all parties (45).
5. Although informed consent is a contract, it is not an agreement to be injured (45). Nor does informed consent imply that lawsuits are impossible. The informed consent document indicates that the athlete has agreed to participate based on understanding the risks involved.
6. At some point the athlete, parent/guardian must sign the informed consent document and thereby agree to participate or allow the underage child to participate in the activity. The signing of the document indicates that both parties are aware of the risks and choose to participate in spite of the risks. These documents must remain in a file in the event the participant is ever injured. No one should be allowed to participate without an informed consent document on file (45).

7. Allow time and provide opportunities for participants and parents/guardians to ask questions. It is a mistake to simply hand out the documents, request signatures, and then collect them.

Pre-Participation Screening and Clearance

A physical examination is imperative for all athletes prior to participating in strength training and conditioning programs. This should include a comprehensive health and immunization history as well as a relevant physical exam, part of which includes an orthopedic evaluation (12,13,19,30). Some type of cardiovascular screening is also recommended (3,13,18,20). Strength training and conditioning coaches do not need a copy of the results, but must require a signed statement verifying proof of medical clearance for athletes to participate.

Warnings

1. Warn in advance of all potential hazards in the strength training and conditioning facility.
2. Document that parents were warned of the hazards of participation before beginning. This usually occurs via the consent/assent document.
3. “Failure to Warn” is an almost universal allegation in sports lawsuits. These allegations are almost impossible to defend against and no proof of negligence or wrongdoing is required (45).
4. Warnings are necessary because the athlete is usually the least informed party in the physical activity. Thus, the coach or supervisor carries the burden of providing the warning(s).
5. Warnings Guidelines:
 - Warnings should be written and cover the full range of potential risks involved
 - Warnings should indicate the hazards and the potential harm that may result from participation
 - Warnings should be understandable by the layperson
 - Warnings should be specific to the particular activity
 - Warnings should not assume that anything is obvious to the athlete or his/her parents/guardians
 - Warnings must cover all aspects of the activity from skills, to transportation, to locker rooms, and so forth
 - Warnings should include the potential consequences and severity of injuries, both short- and long-term
 - Warnings should define and explain terms such as quadriplegia, sickle cell trait, sudden cardiac death, concussion, hypertrophic cardiac myopathy, paraplegia, paralysis, disability, musculoskeletal injury, sprain, strain, arthritis, pathogen, HIV-AIDS, and overuse

- Warnings should include all safety rules and policies
- Warnings should include information about the role of parents, siblings, and significant others
- Verbal and written warnings should be documented and provided consistently and often (25)
- Written records should include warning dates and times, content, and who received the warnings
- Warnings signage should be plentiful, easy to read, and attention should be drawn to the signage whenever appropriate (updates and new policies should be written and provided to all athletes and significant others)
- Warnings are commonly delivered during orientation meetings or initial tours of the facility
- Do not describe warnings as something like, “Well, we have to do this warning stuff, but don’t worry, you’re perfectly safe.” Take warnings seriously, provide them often, and ensure that people understand and appreciate the warnings (33)

Supervision

Supervision is defined as “overseeing the activities of participants,” (45). Estimates have indicated that approximately 80% of athletic injuries involve a lapse in supervision and/or instruction (5,6,37). One of the primary tasks of supervision is to ensure that safety practices are implemented by all concerned (7,8). In short, supervision is designed to prevent injuries due to inappropriate behaviors such as horseplay, daredevil-type stunts, and inattentiveness (31,32).

There are two basic types of supervision, direct and indirect, based on proximity and involvement of the supervisor (23,27,28,38). Direct supervision involves closer contact and direction of the athlete(s) such as in teaching. Usually skill instruction and spotting involve direct supervision. Indirect supervision is common for observing activities in the weight room, but the supervisor is less close to the athletes. In situations when the skills of the activities are well learned, and the athletes are fully competent at performing on their own, then indirect supervision is all that is necessary to maintain vigilant protection of the athletes.

Supervision is an “elastic” concept implying that an experienced strength training and conditioning coach will move easily from direct to indirect supervision, and back, as the situation or the participant’s needs dictate. However, anyone charged with either type of supervision should station him/herself to see all of the activities under his/her jurisdiction. As such, the supervisor should be able to see and stop all activity whenever circumstances dictate (41). Supervision is required as long as athletes are in the facility and sometimes when they are moving from one area of a larger facility to the weight room and back (9). The strength training and conditioning coach should know that his/her responsibility extends to ensuring that athletes moving

from one facility to another are supervised at all times. Be aware that in school settings you may need to collect students at their classroom and return them to their classroom. Do not expect that students, especially young students, will get safely from one place, or building, to another without competent supervision in place (9).

Gender sensitive supervision – The strength training and conditioning coach may be supervising and/or instructing athletes of both genders simultaneously. Obviously, locker room and restroom facilities should be supervised carefully when an instructor or supervisor of the opposite gender must intervene. If possible, have supervisors of both genders present during strength training and conditioning activities.

Stranded participants – Perhaps too often, young participants arrive at the facility via a parent or non-custodial parent (e.g., the parent who drove the carpool). However, if an athlete’s ride home is late when training is completed, supervisors should never be alone with a single youngster in the facility; this is particularly true when the youngster is of the opposite gender. If this should occur, the supervisor should notify local police and have the child picked up by a police officer and transported to the local police station from where pick up of the youngster occurs. Parents/guardians should be aware of this policy from the outset, so that they understand the scope, depth, and duration of the supervisor’s responsibilities. A list of non-custodial parents, relatives, and others who are permitted to transport a youngster should be kept in the facility and non-approved people should not be allowed to transport a youngster without expressed parental or guardian permission. Pre-planning for these types of problems via participation forms can save a great deal of indecision and liability exposure.

Emergency supervision – Emergency plans should be developed and practiced for critical local emergencies such as a serious injury, weather-related threats, and others. Policies and procedures for unusual and threatening situations should be developed, written, communicated to all involved, and practiced. Emergencies may occur when an athlete becomes seriously injured or ill and the supervisor must attend to the injury or illness while awaiting emergency transport. In this type of situation, develop a plan for ensuring the supervision and safety of the other athletes so that they are not unsupervised.

Matching opponents – Strength training and conditioning athletes should be paired or grouped such that they have similar physical characteristics. Partner exercises along with competitive races or other activities can be powerful motivators for training. However, pairing an athlete that is physically more mature with an athlete that is physically less mature can invite problems. A less mature athlete may not even be able to lift the weight needed by the partner performing a selected exercise. Be sensitive to keeping athletes of similar characteristics together so that they can easily assist each other in spotting, moving weights, racking and unracking the bar, and so forth.

Facility

Location

- Ideally, strength training and conditioning rooms should be located on the ground floor with an easily accessible entrance for moving large pieces of equipment in or out

Access

- The room should have clear, unobstructed doors and pathways both inside and out
- All door or floor-to-floor level thresholds should be flush, level, and even
- All exits must be clearly marked, never obstructed, and visible from any location within the facility (special signage for the visually impaired is also required)
- During closed hours, unsupervised access to the room should be prohibited

Ceiling

- Ceiling height should be a minimum of 12 – 14 ft (this height includes other items such as lighting, pipes, fans, and signage)
- If medicine ball training is going to be included, ensure adequate ceiling height and solid walls

Flooring

- The two most common materials for flooring are carpet and various types of rubber flooring
- If carpet is used in the aerobic or warm-up area, it should be treated with antifungal, antiseptic, and antibacterial agents
- In places where free weights touch the ground, facilities should have rubber flooring; otherwise, carpet is recommended for sound and aesthetics
- Provide a firm but resilient floor for plyometric exercises and turf for sprint and agility work

Lighting

- A weight room should be lit to 75 – 100 foot candles (806 – 1,076 lux), and at least 50 foot candles (538 lux) at floor level
- In comparison, a coach's office is often lighted at approximately 50 foot candles
- Lighting should not produce marked shadows

Windows

- Windows should be located a minimum of 20 in. above the floor to protect from rolling bars and plates that could break them
- The floor plan should not have the spotters and lifters near or contacting windows to prevent collision and breakage

Temperature

- The recommended temperature for a facility is 68 – 78°F, (68 – 72°F for summer months and 72 – 78°F for winter months)
- Zone heating is recommended due to its ability to regulate temperatures based on the number of athletes in the room at any one time

Sound

- Music is often used as a motivator while training and the volume should never exceed 90dB

Electrical

- Grounded outlets (GFCI) should be used at all times throughout the room and plenty of outlets should be installed to accommodate all electronic equipment (cleaning machines also need access to electrical outlets)

Signage

- Signage can assist athletes in emergency procedures, operational policies, rules, and safety guidelines

Other considerations

- Drinking fountains should be placed throughout the facility with easy access (however, drinking fountains should not hinder traffic flow or cause a distraction)
- Restrooms for both males and females should be clearly marked and located close to the weight room or in the entrance of the weight room
- Telephones are needed in the strength training and conditioning office for normal use as well as in case of emergencies (an additional line can be placed in the weight room for emergency purposes only)
- Telephones should be located 4 ft off the ground to accommodate athletes with disabilities

In addition to the previous list, there are a few caveats for the safe implementation of a strength training and conditioning program. First, strength training and conditioning activities should be planned, and the required number of qualified staff should be present. Second, recommended guidelines should be followed during peak usage times for minimum average floor space allowance per participant (100 ft²), coach-to-athlete ratios (1:10 junior high school, 1:15 high school, 1:20 college), and number of participants per barbell or training station (up to 3). In ideal circumstances, this corresponds to one strength training and conditioning coach per 3 – 4 training stations and/or 1,000 ft² (junior high school); 5 training stations and/or 1,500 ft² (high school); or 6 – 7 training stations and/or 2,000 ft² (college), respectively. Professional discretion may adjust these guidelines with respect to the practical considerations discussed above.

Performance Safety Team

Modern strength training and conditioning has become a complex professional area with branches that interact with virtually all facets of sports and exercise activities. The literature, consensus statements, position statements, and personal opinions sometimes vary widely and require interpretation by specialist professionals for implementation by the strength training and conditioning coach. Most sports medicine and science specialists will likely agree with the fact that it is almost impossible for them to keep up with the newest literature in their own academic areas. Thus, a coach is unlikely to know what specialists know, and should learn to ask questions and involve specialists in areas of concern that arise in their coaching and athletes (10,15,17,22,24, 26,29,35,39,43,45,46). The performance safety team will also assist the coach in staying up-to-date with the latest trends and discoveries in strength training and conditioning, and the science that underlies the practice. The coach and/or program should seek the following specialists to help:

- Team physician
- Team athletic trainer
- Nutritionist/Dietitian
- Team psychologist/counselor
- Student or other manager
- Others, as needed

Preventing Sudden Death

The Inter-Association Task Force for Preventing Sudden Death in Collegiate Conditioning Sessions published 10 recommendations in January 2012 for the best practices in preventing sudden death. These recommendations were endorsed by the National Strength and Conditioning Association, American College of Sports Medicine, American Medical Society for Sports Medicine, American Osteopathic Academy of Sports Medicine, Canadian Athletic Therapists' Association, Collegiate Strength and Conditioning Coaches Association, and National Athletic Trainers' Association, to name a few. The following are 10 recommendations for preventing sudden death in collegiate conditioning sessions (14):

1. Acclimatize Progressively for Utmost Safety
2. Introduce New Conditioning Activities Gradually
3. Do Not Use Exercise and Conditioning Activities as Punishment
4. Ensure Proper Education, Experience, and Credentialing of Strength Training and Conditioning Coaches
 - A. Education
 - B. Experience
 - C. Credentials

5. Provide Appropriate Medical Coverage
6. Develop and Practice Emergency Action Plan
7. Be Cognizant of Medical Conditions
 - A. Exertional sickling and SCT-related concerns
 - B. Exertional heat stroke
 - C. Cardiac conditions
8. Administer Strength and Conditioning Programs
9. Partner with Recognized Professional Organizations
10. Provide Adequate Continuing Education for the Entire Coaching and Medical Teams

Special Considerations

Recent evidence and medical opinion have arisen to identify several important threats to athlete health of which everyone in athletics should be aware. These threats include sickle cell trait, sudden cardiac death, concussion, exertional rhabdomyolysis, and hyperthermia.

Sickle Cell Trait

Sickle cell trait refers to the change in shape of red blood cells from their typical, rounded frisbee-like shape to a sickle shape ("sickle" refers to the quarter moon shape of the red blood cell) (1,2,4). Sickle cell trait can be fatal. Athletes with sickle cell trait should be allowed to participate in sports. Sickle cell trait diagnosis should have been made at birth, and records of this disease, or simply notification of its presence, should be available to the strength training and conditioning coach. Complications from sickle cell trait occur in many competitive sports. Athletes with this condition will show moderate fatigue, but can still talk and move (1,2,4). If they are suffering from the condition, they will likely have ischemic (lack of oxygen making it to body tissues) pain and muscle weakness (1,2,4). Heat, altitude exposure, asthma, and dehydration may be contributors to the condition even when the exercise is mild. Evidence has shown that incidents occur in hard training rather than games, but the cause or causes of athlete collapse are not yet known. Recommendations include:

1. Slow build-up of exercise intensity
2. Allow athlete to set own pace
3. Provide adequate rest, and when in doubt rest
4. Exclude athletes from pre-season screening fitness tests
5. Stop activity if pain, unusual muscular weakness, or shortness of breath occurs
6. Stay hydrated, stay cool
7. Avoid high caffeine and energy drinks
8. Manage asthma

9. Beware of exercise at higher altitudes, and under high environmental pressures (e.g., scuba diving)
10. Abstain from training when ill
11. Stay in shape year round so that sudden increases in fitness are unnecessary
12. Do not hesitate to seek medical help

Sudden Cardiac Death

Although sudden cardiac death is rare, the devastating consequences obviously merit focused attention. Sudden cardiac death is considered a non-traumatic and unexpected death due to cardiac causes within one hour of symptoms onset (30). Hypertrophic cardiomyopathy is the main cause of sudden cardiac death accounting for 48% of recorded deaths in a sample of 134 athletes, although 17 different causes have been identified (30,36). Unfortunately, most athletes who suffer sudden cardiac death have no symptoms before death and no history of family or personal cardiac problems. Moreover, pre-participation screening using electrocardiography and echocardiography identify only a small percentage of athletes who may have serious cardiac conditions. Thus, the cost of the screening may not be economically feasible given the relatively small positive findings among a very large population of athletes.

Concussion

A concussion is a brain injury caused by a blow to the head that results in shaking, or otherwise traumatizing, the brain. The brain, housed inside the skull, is surrounded by and floats in cerebrospinal fluid. Concussions, although always a part of contact sports, appear to be trending upward based on epidemiological data (11). Concussions do not have to be present with unconsciousness, and most people will fully recover from a concussion. Some serious concussions and concussive events that happen repeatedly may require surgery. Concussions are serious and can be trauma of such significance that the injury affects the athlete for the rest of his/her life (16,42,44,47). Symptoms of a concussion include:

- Unconsciousness
- Unclear thinking
- Inability to concentrate
- Forgetfulness
- Headache
- Blurred vision
- Nausea and vomiting
- Dizziness
- Light sensitivity

- Ataxia (balance problems)
- Agitation
- Inability to sleep

If an athlete suffers a concussion, the athlete should be removed from play for the entire day. The seriousness of this type of head injury cannot be judged by a layperson. Appearance of any of the symptoms listed above should merit a medical examination. The athlete should be removed from all sport participation until cleared by a physician.

Exertional Rhabdomyolysis

Exertional rhabdomyolysis is the degeneration or destruction of muscle tissue because of excessive and unaccustomed exercise. Rhabdomyolysis has been seen in emergency medical settings due to circumstances such as suspension syndrome when someone is held aloft for period of minutes to hours while blood flow to and from the legs is cut off due to a harness or rope acting like a tourniquet. In strength training and conditioning settings, exertional rhabdomyolysis is caused by excessive exercise (21,34,40). The best treatment for rhabdomyolysis is prevention. Guidelines to avoid rhabdomyolysis include:

1. Do not begin training programs with excessive and intense exercise (instead, build training loads progressively)
2. Separate freshmen from upper classmen due to varying fitness levels
3. Maintain adequate hydration
4. Feelings of dizziness should be taken seriously by reducing training intensity
5. Train progressively and use delayed onset muscle soreness (DOMS) as a clue that training may be too difficult for the athletes

Symptoms include:

- Pain in the muscle(s), and on exertion
- Muscle weakness and swelling
- Myoglobin in the urine (myoglobin turns the urine a brown color)
- Other muscle cell proteins and enzymes are found in the blood
- Increased amount of large proteins in the blood (can lead to kidney failure and death)

Hyperthermia

Hyperthermia means overheating. Hyperthermia is a potentially fatal condition, most often seen in football during pre-season training during the latter summer months when outside temperatures are hot, accompanied by high humidity. Heat injury can range from mild heat cramps to heat stroke. Unfortunately, a number of athletes have died from heat stroke during training. Heat stroke is completely preventable, though. Recommendations for avoiding heat-related performance problems include:

1. Encourage and even enforce frequent rest periods in hot and humid environments
2. Encourage maintenance of hydration by frequent drink breaks (thirst is a poor indicator of dehydration—encourage fluid intake before the athlete is thirsty)
3. Urine color can be a helpful tool in gauging hydration (dark yellow or brown urine indicates dehydration—urine should be light yellow in color or clear)
4. Compare pre-practice weight with after practice weight and try to replace about 1.5 times the weight difference (i.e., loss) in fluids following practice
5. Acclimatize to the high heat and humidity by progressing training very slowly
6. Use the cooler parts of the day to train
7. Wear light clothing and light colors
8. Do not be afraid to stop training if the environment is prohibitive for safety

Factors include:

- Environmental temperature
- Environmental wind direction and speed (wind is a potential cooling agent)
- Environmental humidity (greater than 60% can be problematic, especially when combined with high temperatures)
- Dark clothing tends to absorb heat from the sun
- Fitness and acclimatization
- Age (children do not adapt to high heat and humidity as rapidly as adults)
- Hydration (the athlete must remain hydrated throughout training)
- Over-fat (fat is an insulating tissue and athletes who are over-fat are more predisposed to heat injury than leaner athletes)
- Sick cell trait is aggravated by hot environments

In closing this chapter, always keep the best interest of the athlete foremost in your mind. Training decisions should always proceed from a safety premise. Remember, that sensible strength training and conditioning is injury prevention. However, like all great things, strength training and conditioning can be abused and thereby obtain the opposite of safety. Well-designed training, well-instructed and well-supervised athletes do not tend to be injured or suffer from overtraining. When in doubt, rest. When training, be ever vigilant and observant of technique errors. Be sure to pay attention to detail so that the small errors, that by themselves may not merit much attention, do not lead to bigger errors.

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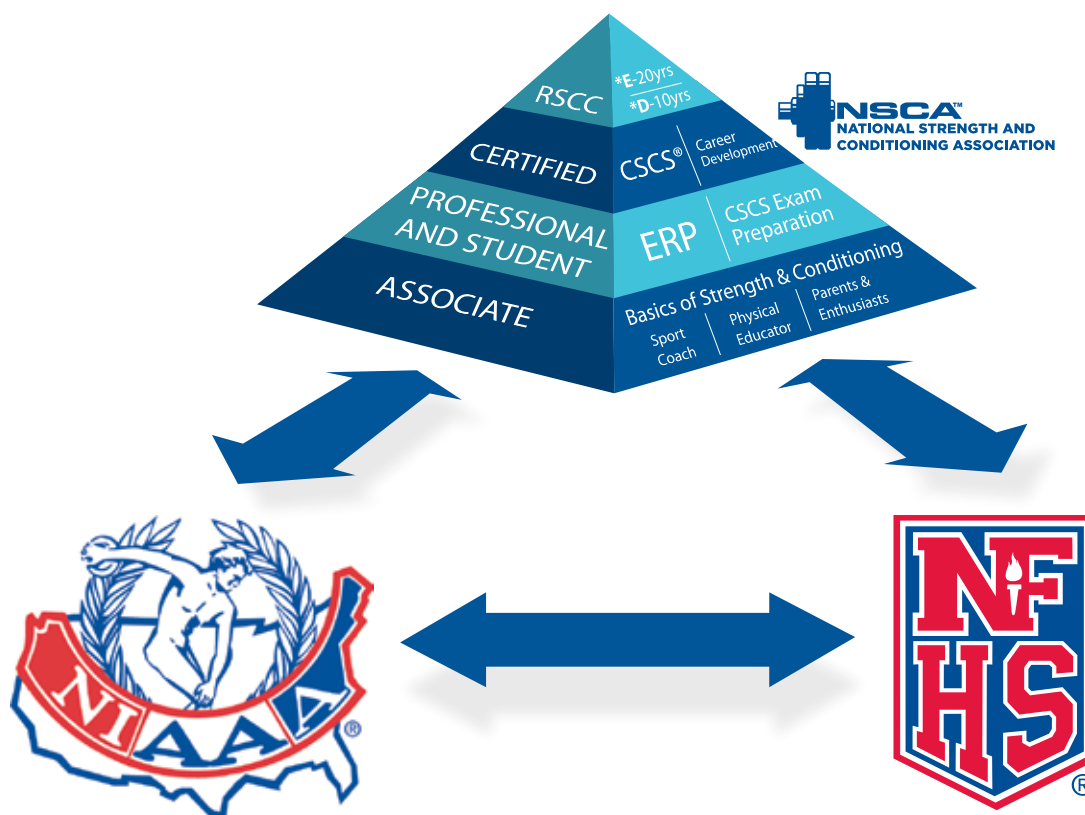
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